Robust Contactless Palmprint Verification with Deep Metric Learning: Checkpoint 1 — Data Foundation & Initial Modeling

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***Abstract*—Contactless palmprint verification offers a hygienic and user-friendly biometric modality, but real-world capture variations can severely degrade system performance. This checkpoint report introduces our verification system built on the PolyU–IITD contactless palmprint dataset (611 subjects, 12,220 images) with a strict subject-disjoint evaluation protocol. We describe the data engineering pipeline, present exploratory data analysis, and detail our initial model architecture—ResNet18 with ArcFace angular margin loss. Training convergence analysis shows the model reaches stable performance by epoch 32, with validation EER of 1.13%. Full test-set evaluation is deferred to Checkpoint 2.**

I. Problem Statement

Contactless palmprint verification is an emerging biometric modality that captures palm images without physical contact, improving hygiene and user acceptance compared to fingerprint sensors. Palmprints contain rich discriminative features—principal lines, wrinkles, and texture patterns—that remain stable over time, making them well-suited for identity verification [4], [5].

However, contactless capture introduces challenges absent in contact-based systems. Variations in hand pose, sensor distance, ambient lighting, motion blur, and compression artifacts all affect image quality. A practical verification system must not only achieve high accuracy on clean images but also degrade gracefully under these real-world conditions. When performance degrades, both security (increased false accepts) and usability (increased false rejects) are compromised.

This project addresses the question: *How robust is a deep learning-based palmprint verification system to common real-world capture variations, and which factors pose the greatest risk to deployment?*

A. Project Objectives

1) Train a deep metric learning model for palmprint verification with subject-disjoint evaluation.
2) Establish baseline performance on clean data using standard biometric metrics (EER, FAR/FRR).
3) Benchmark robustness under systematic image corruptions spanning noise, blur, geometric, and compression categories.
4) Identify the most damaging degradation factors to guide future mitigation strategies.

II. Data Engineering

A. Dataset

We use the **PolyU–IITD Contactless Palmprint Database v3.0** [1], comprising 611 subjects with 20 images each (10 left hand + 10 right hand), totaling 12,220 images. All images are 128×128 grayscale ROI-cropped palmprint regions.

B. Preprocessing Pipeline

Each grayscale image is converted to 3-channel RGB for compatibility with ImageNet-pretrained backbones. Images are normalized using ImageNet channel statistics ($\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$). During training, we apply data augmentation including random cropping, horizontal flipping, rotation ($\pm 15°$), color jitter, Gaussian blur, and random erasing to improve generalization.

C. Subject-Disjoint Splits

To prevent identity leakage, we use subject-disjoint splits where no subject appears in more than one partition. Each (subject, hand) combination is treated as a unique class, doubling the effective number of identities:

TABLE I
Dataset splits with subject-disjoint protocol.

Split	Subjects	Images	Classes
Train	427 (70%)	8,540	854
Validation	61 (10%)	1,220	122
Test	123 (20%)	2,460	246

A quality audit identified 4 empty placeholder subjects (H_ID608–611) in the original dataset; these were excluded from all splits.

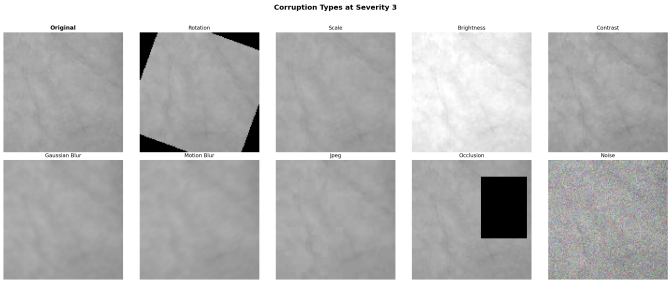

Fig. 1. Original palmprint image and 9 corruption types at severity level 3. Top row: Original, Rotation, Scale, Brightness, Contrast. Bottom row: Gaussian Blur, Motion Blur, JPEG, Occlusion, Noise.

III. Exploratory Data Analysis

A. Class Balance

The dataset is perfectly balanced by design: each subject contributes exactly 10 images per hand, yielding uniform class sizes across all splits. This eliminates class imbalance as a confound in our evaluation. With 854 training classes each containing 10 samples, the model receives consistent supervision across identities.

B. Sample Visualization

The 128×128 ROI-cropped images capture the central palm region containing principal lines, wrinkles, and texture patterns. Visual inspection confirms consistent quality across subjects, with clear visibility of discriminative features. Inter-subject variation in line patterns provides the basis for identity discrimination, while intra-subject consistency (across the 10 captures per hand) supports reliable matching.

C. Corruption Types Overview

To motivate our planned robustness evaluation, we preview the 9 corruption types that will be applied at 5 severity levels each. Fig. 1 illustrates these corruptions at severity level 3, spanning noise (Gaussian), blur (Gaussian, motion), geometric (rotation, scale), photometric (brightness, contrast), compression (JPEG), and occlusion categories. These corruptions model real-world capture degradations that a deployed system would encounter.

IV. Baseline Architecture and Training

A. Architecture

Our model uses a two-stage architecture with approximately 12M parameters:

1) **Backbone:** ResNet18 [3] pretrained on ImageNet, with the final fully-connected layer replaced by an identity mapping, yielding 512-D feature vectors.
2) **Projection Head:** A two-layer MLP ($512 \to 512 \to 256$) with batch normalization and ReLU, producing L2-normalized 256-D embeddings on the unit hypersphere.

During training, an ArcFace [2] classification head computes $\text{logit}_j = s \cdot \cos(\theta_j + m \cdot \mathbb{1}[j = y])$ with scale $s = 30$ and angular margin $m = 0.3$. The ArcFace head is discarded at inference; only the embedder is used for verification via cosine similarity.

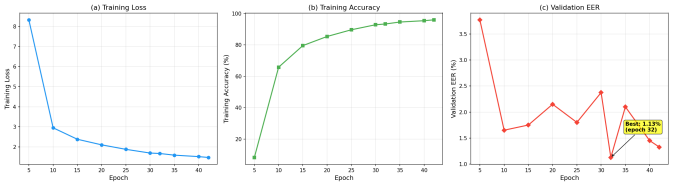

Fig. 2. Training convergence: (a) loss decreases from 8.3 to 1.5, (b) accuracy rises to 96%, (c) validation EER reaches minimum 1.13% at epoch 32.

B. Training Configuration

The model is trained with AdamW (lr = 3×10^{-4}, weight decay = 10^{-3}) and a cosine learning rate schedule for 50 epochs with early stopping. The best checkpoint is selected based on validation EER.

C. Training Convergence

Fig. 2 shows the training convergence over 42 epochs (early stopping triggered). Key observations:

- Training loss decreases from 8.3 to 1.5, indicating effective optimization.
- Training accuracy rises to 96%, confirming the model learns to discriminate among 854 identities.
- Validation EER reaches a minimum of 1.13% at epoch 32, which is selected as the best checkpoint.

The smooth convergence and low validation EER suggest the model has learned a discriminative embedding space. Full test-set evaluation with verification pair generation and robustness benchmarking will be presented in Checkpoint 2.

V. Code Repository

The complete source code is available at: `https://github.com/onggiahuy97/cs228-palmprint`.

Repository structure:

- `src/` – Training, evaluation, and data loading scripts
- `splits/` – Subject-disjoint split definitions (JSON)
- `figures/` – Generated plots and visualizations
- `checkpoints/` – Model weights
- `requirements.txt` – Python dependencies
- `README.md` – Setup and usage instructions

Key files:

- `src/train.py` – Training loop with ArcFace loss and early stopping
- `src/dataset.py` – Dataset class with augmentation pipeline
- `src/model.py` – ResNet18 backbone + projection head
- `src/create_splits.py` – Subject-disjoint split generation

REFERENCES

[1] PolyU–IITD Contactless Palmprint Database v3.0, The Hong Kong Polytechnic University.

[2] J. Deng, J. Guo, N. Xue, and S. Zafeiriou, "ArcFace: Additive angular margin loss for deep face recognition," in *Proc. IEEE/CVF CVPR*, 2019, pp. 4690–4699.

[3] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE CVPR*, 2016, pp. 770–778.

[4] D. Zhong, X. Du, and K. Zhong, "Decade progress of palmprint recognition: A brief survey," *Neurocomputing*, vol. 328, pp. 16–28, 2019.

[5] D. Zhang, W.-K. Kong, J. You, and M. Wong, "Online palmprint identification," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 25, no. 9, pp. 1041–1050, 2003.